

Maximizing shared benefits in renewable energy communities: A Bilevel optimization model

Virginia Casella, Giulio Ferro, Luca Parodi^{*}, Michela Robba

Department of Informatics, Bioengineering, Robotics and Systems Engineering (DIBRIS), University of Genoa, Genoa, Italy

HIGHLIGHTS

- Bi-level model for optimal management of an Energy Community (EC) is presented.
- EC participants minimize costs, while the EC manager maximizes total shared energy.
- The new Italian incentive scheme is approximated using two different techniques.
- Impact of EC participation analyzed, showing real benefits for participants.

ABSTRACT

To respond to the global need for sustainable energy solutions and the imperative to combat climate change, Renewable Energy Communities (REC) have emerged as a promising solution to achieve energy transition goals. Of course, some optimization tools need to be developed to face the challenges related to their operational management and maximize their potential. In this context, this paper proposes a bilevel optimization approach for the optimal management of a REC, focusing on maximizing shared energy and economic benefits. The high-level models the problem of the Energy Community Manager (ECM), who aims at maximizing shared energy rewarded with incentives depending on the plants according to the new legislation; instead, the low-level problems focus on each Energy Community Participant (ECP) aiming to minimize individual costs. To solve this problem Karush-Kuhn-Tucker (KKT) conditions are exploited to convert low-level problems into constraints for the high-level problem. Two different approaches (MILP and NLP formulations) to approximate the high-level objective function are proposed and tested, and the best approach is applied to a case study involving ten ECPs. The scalability of the proposed approach is evaluated as well as the impact of the most influencing parameters. According to the results, each ECP would obtain an annual income for sharing energy, which could be significant, especially when proper pricing strategies are considered. Moreover, the proposed model is suitable for online operations as the runtime is quite low.

1. Introduction

In response to the escalating environmental crisis and the necessity to mitigate climate change, exploiting Renewable Energy Sources (RES) and developing sustainable energy solutions have been globally incentivized. In this framework, Renewable Energy Communities (REC) have emerged as a promising paradigm in the energy transition. Indeed, only renewable plants are allowed in an energy community by law and local energy consumption is encouraged, significantly reducing the stress on the main grid. Different kinds of ECs can be considered (rural RECs, citizen), and in this paper, attention is focused on the regulation and incentives of RECs that will be referred to as ECs through the paper. In particular, the definition of EC has been suggested in [1]. However, since the legal definition can vary between jurisdictions, it's essential to refer to the specific legal framework of the country. In general, an EC is defined by its collaborative approach to energy generation,

consumption, and management; indeed, it involves the active participation of all members, emphasizing shared responsibility and communal benefits.

Since an EC is a complex entity from legislative, juridical and social perspectives, a figure coordinating all members can make the community more efficient in several ways. This figure, who will be referred to as 'Energy Community Manager' (ECM), must first and foremost manage regulatory compliance by acting as an intermediary with authorities and facilitating communication among the community members by, for example, overseeing how to split the incentive that the EC obtains based on the amount of energy shared as it will be explained later. Therefore, the manager does not need any particular expertise or qualifications, also because this role is not required by law, other than the will to carry forward the concepts of the energy transition. Nevertheless, it must be considered that most of the time community members have limited expertise in the energy sector; hence, to ensure operational efficiency, optimize the use of energy resources and oversee technical aspects and

^{*} Corresponding author.

E-mail address: luca.parodi@edu.unige.it (L. Parodi).

<https://doi.org/10.1016/j.apenergy.2025.125562>

Received 25 October 2024; Received in revised form 20 January 2025; Accepted 14 February 2025

Available online 28 February 2025

0306-2619/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature			
Acronyms			
DM	Decision Maker	CAP_i^S	[kWh]Capacity of the storage system
EC	Energy Community	$\Gamma_i^{S, ch}$	[/]Charging efficiency of the storage system
ECP	Energy Community Participant	$\bar{X}_i^S$	[/]Maximum state of charge of the storage system
EV	Electric Vehicle	$E_i^{L, flex}$	[kWh]Energy required by flexible of load over the entire optimization horizon
KKT	Karush-Kuhn-Tucker	CAP_i^{EV}	[kWh]Capacity of the electric vehicle
NLP	Non-Linear Programming	$\bar{P}_i^{EV}$	[kW]Maximum power of the electric vehicle
REC	Renewable Energy Community	$\bar{X}_i^{EV}$	[/]Maximum state of charge of the electric vehicle
V2G	Vehicle To Grid	$X_{i,t}^{EV^*}$	[/]Desired value for the EV state of charge
DR	Demand Response	ζ	Parameter affecting the quality of approximation according to the NLP formulation
ECM	Energy Community Manager	T	[h]Length of the optimization horizon
ESS	Energy Storage System	N	[/]Number of REC participants
GSE	Gestore Servizi Elettrici	P	[/]Number of REC participants with a production plant
MILP	Mixed-Integer Linear Programming	$p_{i,t}^{G, out}$	[kW]Power injected into the grid by user i in each time interval
POD	Point Of Delivery	$p_{i,t}^{EV}$	[kW]Charging power of the electric vehicle
RES	Renewable Energy Source	$p_{i,t}^{S, dch}$	[kW]Discharging power of the storage system
Sets		$P_i^{L, flex}$	[kW]Flexible load power consumption
$S^T = \{1, \dots, T\}$	Set of time instants	$x_{i,t}^{EV}$	[/]State of charge of the electric vehicle
$S^N = \{1, \dots, N\}$	Set of energy community participants	C^{sell}	[€/kWh]Costs for selling energy to the grid
$S^P = \{1, \dots, P\}$	Sub-set of participants with a production plant	$\bar{P}_i^S$	[kW]Maximum power of the storage system
Nomenclature		$\Gamma_i^{S, dch}$	[/]Discharging efficiency of the storage system
Δ	[h]Time step	$\bar{X}_i^S$	[/]Maximum state of charge of the storage system
α	[/]Weight in the objective function	$\bar{P}_i^{L, flex}$	[kW]Maximum power that can be delivered to the flexible load
β	[kW]Auxiliary variable β added to avoid the $maxmin(\bullet)$ formulation in the ECM objective function	$\Gamma_i^{EV, ch}$	[/]Charging efficiency of the electric vehicle
$p_{i,t}^{G, in}$	[kW]Power absorbed from the grid by user i in each time interval	$\bar{X}_i^{EV}$	[/]Maximum state of charge of the electric vehicle
$P_{i,t}^{PV}$	[kW]Power from photovoltaic plant	$T_i^{EV^*}$	[h]Time instant at which the desired EV state of charge must be reached
$p_{i,t}^{S, ch}$	[kW]Charging power of the storage system	$TIP_{p,t}$	[€/kWh]REC incentive for each member who has a production plant ($p \in P \subseteq N$)
$P_{i,t}^{L, fix}$	[kW]Fixed load power consumption		
$x_{i,t}^S$	[/]State of charge of the storage system		
C_t^{buy}	[€/kWh]Costs for purchasing energy from the grid		

maintenance, the manager could request an external consultation or use an optimization tool to build a robust organizational structure, foster the long-term success of the energy community and achieve shared goals of sustainability and energy autonomy.

This paper presents a new bilevel architecture for the optimal management of ECs inspired by the approach in [2]. The main reason that motivates a bi-level structure is the presence of different actors: the participants and the manager. The proposed architecture involves many Energy Community Participants (ECPs) that aim to minimize their costs at the low-level. Each ECP is characterized by different features such as renewable plants, electric vehicles, storage systems, and a partially flexible load. The high-level represents the energy community manager who aims to maximize the revenue generated by sharing energy. This paper proposes an optimization problem for an EC, in which each low-level optimization problem is converted into constraints for the high-level problem by using the Karush-Kuhn-Tucker (KKT) conditions. The model considers potentially conflicting objectives, ensuring optimal operations of the ECPs while maximizing shared energy. It must be noted that the incentive scheme considered in the high-level problem is very complex and needs to be approximated; to do that, in this paper, two different formulations are presented: with the two approximation a mixed-integer linear programming (MILP) problem, and a non-linear programming (NLP) problem are obtained.

The proposed approach is tested in a case study where an EC composed of ten ECPs is considered, and the impact of participating in

an EC is discussed. Then, a performance analysis for the two proposed formulations is provided.

The main contributions of this work can be summarized as follows:

- A novel optimization model for the optimal management of a EC is presented. In particular, both the ECPs and the ECM objectives are considered by maximizing the shared energy while minimizing the participants costs.
- The real incentive scheme recently introduced by the Italian legislation is considered in the ECM problem. As this scheme has a very complex formulation, it is approximated using two different techniques. In the case study, the performances of the two formulations are tested.
- An accurate representation of a real EC is presented in the case study. The impact of participating in a EC is analyzed showing the real benefit received by the ECPs.

The remaining part of the paper is divided as follows: [Section 2](#) presents the state of the art. [Section 3](#) presents the considered pricing mechanism for the incentive. [Section 4](#) presents the problem definition, with subsections for low-level and high-level problems. Then, in [Section 5](#) the optimization problem is described. The model is tested on a case study in [Section 6](#), where the results are presented and a performance analysis for the two proposed formulations is performed; moreover, a scalability analysis is conducted to test the model on larger scenarios

Table 1

Literature research summary on RECs in the Italian context.

	Publication year	Problem class	Max{Shared-energy}	Min{ECP costs}	Updated Incentive Scheme [24]
[14]	2022	MILP	✓	✓	–
[16]	2023	MIQCP	✓	–	–
[18]	2022	MILP	✓	–	–
[22]	2024	LP	✓	✓	–
[23]	2024	MILP	✓	✓	–
This paper		MILP/NLP	✓	✓	✓

and the impact of the most influencing parameters is evaluated in two different sensitivity analysis. Finally, Section 7 concludes the paper.

2. State of the art

Like all initiatives for a more sustainable future, ECs also present technical, economic, regulatory, and social challenges [3]. Successful EC projects recognize and address challenges at both planning and management stages to ensure long-term sustainability and efficiency. In this context, modelling techniques and optimization methods [4] play a crucial role as they provide a comprehensive framework to design resilient and adaptive energy systems during the planning stage, while offering real-time solutions for managing the dynamic nature of renewable energy resources in the operational phase. In this context, it is crucial to analyze the state of the art to differentiate between planning challenges, such as the optimal sizing and integration of renewable energy resources [5], and operational management issues, including strategies for energy sharing and the coordination of distributed resources. This distinction is particularly relevant for addressing the complexities of renewable energy communities under evolving legislative frameworks.

Focusing on technical and engineering challenges, planning includes efficient integration of intermittent RES, optimal sizing of renewable production plants and Energy Storage Systems (ESS) considering demand patterns to reach the most effective configuration and enhance the potential energy sharing mechanisms. Indeed, even though photovoltaic technology is confirmed as the one candidate to become the main generation technology for this type of entity due to its characteristics of cost-effectiveness and ductility, technologies such as wind, biomass or small-hydro would instead be crucial to maximizing community energy self-sufficiency [6]. Nonetheless, few papers in literature address this topic. Among them, [7] presents a study where the optimal planning of a EC is determined by applying genetic algorithms in a multi-optimization problem. They consider the prosumer perspective and minimize the costs, the solar excess and obtain a fair benefits distribution. However, since they refer to different legislation, they do not consider the contribution of shared energy. Another interesting work is proposed in [8], where authors provide an optimization model for the optimal sizing of a EC in the Italian context. Moreover, they compare physical with virtual schemes and centralized with decentralized schemes. The Italian scenario is investigated also in [9], where authors propose a multi-criteria optimization procedure to size the active part of an EC by evaluating self-consumption and self-sufficiency of the energy community as key performance indices. The optimal design problem is faced also in [10]: an annualized investment optimization approach considering that one EC member can invest in additional renewable generation (namely PV and wind capacity) is presented to support EC's investment decisions on the renewable generation portfolio and operational electricity sharing management. A Multi-Objective Particle Swarm Optimization algorithm is employed in [11] to determine the optimal sizing for each production and storage unit within ECs regardless of the number of participants; the sizing of the various units considers the leveled cost of energy, the self-consumption ratio, and the self-sufficiency ratio as economic and technical criteria. Authors in [12] develop a data-driven approach for the optimal design of an EC: the size of the renewable

sources is optimized under energy, economic and environmental criteria in a multi-objective framework where green energy use is also maximized.

Management decision problems are instead more interesting and challenging; mainly, they examine how to constantly match demand and production curves to maximize the shared energy and the relative incentive, how to optimize the operation of an ESS for balancing intermittent energy production and reducing peak load, how to coordinate the various distributed energy sources to maximize the overall system efficiency and how to develop effective demand-response strategies to provide flexibility services to the grid. In [13] authors investigate how taking into account economic or technical goals affects the optimal operation of ECs: the two optimizations can be viewed as two extremes: the cost optimization has the best financial outcome for the EC and its members, while the resilience optimization results can be viewed as a worst-case scenario of how much it can cost to use the EC for grid services. The results show that the best solution lies somewhere between these two extremes. Authors in [14] propose a MILP model for the optimal operation of an energy community maximizing a social welfare term. In [15], a bilevel model for optimal management is presented: the low-level problem (MILP) models the market-clearing task considering a social welfare term. In contrast, the high-level problem prioritizes profit redistribution by maximizing the smallest (absolute or relative) profit increase that the entities experience when participating in the community. Authors in [16] propose a model to optimally manage an EC through reversible solid oxide cells: EC operations are optimized by maximizing the overall incentives and minimizing the aggregated cost for energy exchange with the grid. In [17], a three-module strategy for scheduling an ESS in EC is proposed to optimize economic revenues and minimize operation costs. Authors in [18] present a stochastic MPC to optimize EC operations with a proper control strategy of ESS by adopting a Time Delay Neural Network to forecast the future values of the energy features in the community. In [19], authors propose a hierarchical energy-sharing mechanism where end-users and producers perform a self-scheduling to minimize costs; then, the EC manager optimizes the energy trades with the external suppliers and then performs a sharing mechanism based on three different energy classes. The energy sharing model proposed in [20] exploits the Shapley Value to distribute the community-owned renewable energy among the members considering the individual participation of each member in demand response events, launched to promote the balance between consumption and generation inside the energy community; according to the results, the application of the DR programs allows to reduce the community energy cost of 12.86 % and the members saw their energy costs reduced on average by 31.03 %. Authors in [21] propose a methodology to optimize energy management in prosumer energy communities with the primal focus on maximizing self-consumption and self-sufficiency within the community by managing prosumers' batteries for community purposes. In [22], authors propose a cooperative distributed optimization framework for Italian energy communities; according to the results, this approach can yield financial benefits of 14.53 % using the shared energy concept. In [23] a centralized mixed-integer linear programming optimization scheme is proposed to optimally coordinate prosumers' Battery Energy Storage Systems and Thermostatically Controlled Loads to reduce the overall energy cost for the community considering cost

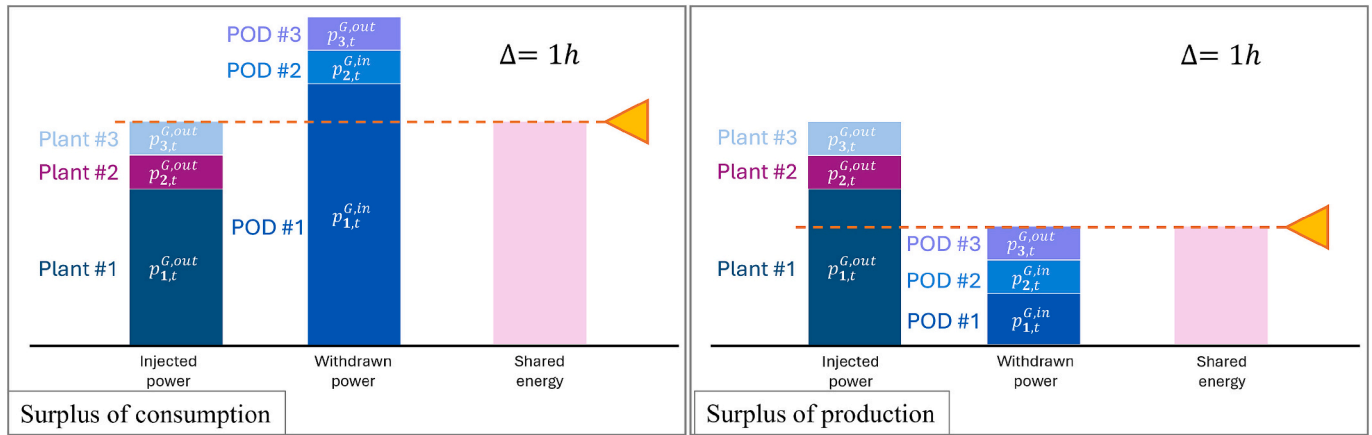


Fig. 1. Graphical representation of the shared energy calculation [27].

minimization and shared energy maximization.

To the best of authors' knowledge, not many papers in the literature that treat the optimal management of ECs consider the real incentive schemes and, in general, ECs are treated as classical energy districts and microgrids. Indeed, this is because the regulations and the incentive mechanisms are still not homogeneous and each country has its own rules. Since different systems cannot be compared, to make the comparison reported in Table 1 more reasonable, only the latest papers related to the optimal management of RECs (according to the definition given in related European directive RED II) in the Italian context are reported. From this table, it is evident that this paper is the only one that introduces in new incentive according to the newest legislation [24].

Indeed, this paper considers an incentive mechanism related to time varying prices, plant sizes and date of first connection to the grid. This comes from the most recent regulation in some countries (such as Italy). The resulting problem is challenging to solve because the objective function is highly nonlinear. The proposed formalization represents a novelty in the literature dealing with the most recent legislation relevant to ECs.

In the following Section, the overall framework relevant to the incentive scheme is presented.

3. The considered incentive scheme

Since the definitions of ECs included in the European directive (EU) 2018/2001 of 11 December 2018, the so-called RED II, had to be transposed into national regulations by 30 June 2021 and each state freely decides what legal forms to accept as ECs and what incentives to introduce to ensure their economic viability, these realities will have very different characteristics as they move from one border to another [6]. Therefore, it follows that it is important to differentiate from state to state and specify which context is being studied.

In particular, referring to the Italian framework, the most up-to-date legislation [24] by the GSE "Gestore Servizi Elettrici" (Italian energy service operator) [25] dates back to April 2024, proving that ECs are an innovative topic and a hot issue about which there are still open questions to be addressed to maximize their effectiveness. According to [24], a EC must be established as an autonomous legal entity and must include at least two members: a consumer and a production plant, with two distinct connection points that must fall within the area beneath the same primary substation. As legal entity, each EC needs a statute, which specifies that a EC must be: (a) based on open and voluntary participation, autonomous, and effectively controlled by shareholders or members that are located in the proximity of the renewable energy projects that are owned and developed by that legal entity, (b) whose members are natural persons, enterprises or local authorities, including municipalities, and (c) with the primary purpose to provide

environmental, economic or social community benefits for its members or for the local areas where it operates, rather than financial profits. The statute also indicates a representative, who could coincide with the ECM, who is defined as the person to whom technical and administrative management of the EC is entrusted, and as the counterparty of the contract with the GSE for obtaining the benefits provided by the service. All EC participants retain their rights as end customers. As regards production units, any plant powered by RESs can be included in a REC; for example, photovoltaic, hydroelectric, wind, biogas, solid biomass systems are allowed. Storage systems can be included in a REC; how the electricity withdrawn, absorbed and reintroduced from the storage systems is treated, for determining the incentives, is specified in [26].

A key concept when talking about ECs is *shared energy*, which is important from a technical point of view because the greater the shared energy, the greater local consumption and therefore the less stress on the grid. In Italian legislation, shared energy is defined as the "minimum between the electricity globally injected and the electricity globally withdrawn" (1).

$$e_t^{sh} = \min \left\{ \sum_{i \in S^N} p_{i,t}^{G,in}, \sum_{i \in S^N} p_{i,t}^{G,out} \right\} \Delta t \in S^T \quad (1)$$

where $p_{i,t}^{G,in}$ [kW] and $p_{i,t}^{G,out}$ [kW] are the power flows absorbed from/injected into the grid by each user i in each time interval, which length is defined by Δ [h]. Fig. 1 shows the definition of shared energy for a basic example with three production plants and three Points of Delivery (POD) in two different cases: surplus of consumption and surplus of production. From the figure, it is evident that the minimum is always taken.

This quantity is calculated hourly and according to an ex-post calculation based on the members' smart meters. Electricity sharing occurs using the existing distribution network, and all energy exchanges are virtual, meaning there is no electrical grid modelling like in microgrids or sustainable energy districts.

Shared energy is also important from an economic perspective, as the EC obtains some incentives based on how much energy is shared. Specifically, three economic contributions are admitted: a) incentive for shared electricity, b) valorization of self-consumed electricity and c) withdrawal of electricity fed into the grid by the GSE, where requested.

As far as the incentive at point a) is concerned, the equation provided by the legislation is the following:

$$TIP_{i,t} = \{ \min [CAP_i; TP_i^{base} + \max(0, 180 - P_{z,t})] + FC^{zone} \} \cdot (1 - F) \quad (2)$$

$i \in S^N, t \in S^T$

This equation shows that the incentive is made up of a fixed and a variable part: the fixed part TP_i^{base} varies according to the size of each

Table II

Base tariff and cap depending on the plant size.

Size of the plant	TP_{base}	CAP
$P < 200$ kW	0.08 €/kWh	0.10 €/kWh
$200 \text{ kW} < P < 600$ kW	0.07 €/kWh	0.11 €/kWh
$P > 600$ kW	0.06 €/kWh	0.12 €/kWh

Table IIICorrection factor for PV productivity FC_{zone} .

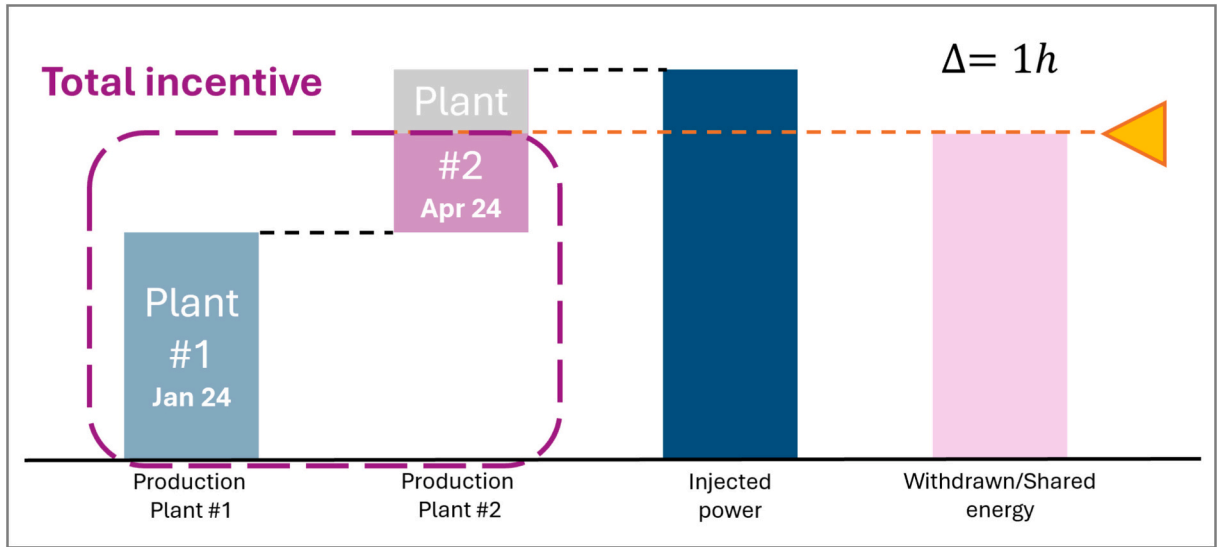
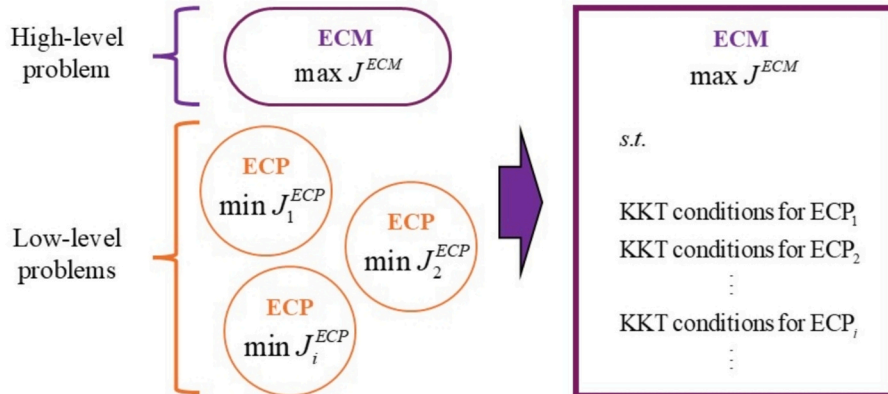
Central Italy	+0.004 €/kWh
Northern Italy	+0.010 €/kWh

plant i (Table II), the variable part according to the hourly market price of energy P_{z_t} . The fixed part decreases as the size of the plants increases, while the variable part fluctuates depending on the energy price (as the energy market price decreases, the variable part increases). For the incentive an upper bound is defined: CAP_i is the threshold value that the tariff can assume and is defined based on the size of each plant (Table II). Then, to consider the lower productivity of photovoltaic systems in central-northern regions with respect to southern ones, the correction factor FC_{zone} has been introduced; in this way, for central-northern regions the tariff is increased as reported in Table III. Finally, parameter F

is linked to the possible payout of a capital contribution: in most cases, it varies linearly between 0, in the case in which no capital contribution is foreseen, and 0.5, in the case of a capital contribution equal to 40 % of the investment. For more information about this, please refer to Part III of [24].

The calculation of the overall incentive requires further investigation; indeed, as presented above, each production plant has an incentive depending on its size. However, since shared energy can collect different contributions from the production plants, the overall incentive depends on which plants are considered in the definition of the shared energy. In fact, as presented in the right side of Fig. 1, some plants or a portion of them can be excluded. The rule coming from the legislation is that the plants are ordered according to the date of the first connection to the grid. In Fig. 2, an example about how the tariff is calculated in case of production surplus is reported. First, the energy produced by plant #1 connected to the grid in January 2024 is counted, then plant #2 connected to the grid in April 2024 receives an incentive not on the total produced energy but on how much it contributed to the energy sharing. In this way, each plant receives the incentive based on how much it actually contributes to energy sharing and the EC obtains an incentive equal to the sum of the incentives obtained by each plant.

Regarding the valorization in point b), the GSE determines, based on the self-consumed electricity, the ARERA valorization fee, which varies every year (in 2023 it was equal to 8.48 €/MWh). Finally, all renewable

**Fig. 2.** Tariff calculation in case of surplus of production [27].**Fig. 3.** Schematic representation of the proposed bi-level approach.

electricity produced but not self-consumed remains available to producers and is valued at market conditions; for this energy it is possible to request access to the economic conditions of dedicated withdrawal from the GSE – point c).

4. Decision problem definition

As the EC framework needs to consider two different perspectives - that of the participants and that of the manager - the proposed architecture consists of two distinct levels. The low-level problems correspond to the ECPs whose goal is to minimize their costs. The high-level problem is an ECM who aims to maximize shared energy, in order to obtain a higher incentive. The mathematical formulation of the two levels is presented in the following; then, in Section 5 the low-level problems are converted into constraints for the high-level problem by introducing KKT conditions. A schematic representation of the proposed approach is presented in Fig. 3.

4.1. Low level: EC participants

The low level of the proposed approach is focused on each EC participant, which aims to minimize its own costs. At this level, one optimization problem for each ECP, denoted with index i , is set and solved. In principle, ECPs can be producers, consumers or both (prosumers). The objective function for each user is composed by two different terms

$$J_i^{ECP} = \Delta \sum_{t \in S^T} (C_t^{buy} p_{i,t}^{G,in} - C_t^{sell} p_{i,t}^{G,out}) + \alpha (x_T^S - x_0^S)^2 \quad i \in S^N \quad (3)$$

The first term is related to grid cost minimization, where C_t^{buy} and C_t^{sell} [€/kWh] are the costs for purchasing/selling energy from/to the grid, respectively; please note that to evaluate costs and benefits properly, the bidirectional power exchange with the main grid is split into two positive terms $p_{i,t}^{G,in}$ and $p_{i,t}^{G,out}$ [kW]. In addition to grid costs, the second term is added to avoid exploiting the ESS 'for free', by minimizing the difference between the ESS state of charge at the end of the optimization horizon (x_T^S) and its value at the beginning of the optimization horizon (x_0^S). Note that, a weighting parameter α is also introduced.

The power balance for each user is defined as

$$p_{i,t}^{G,in} - p_{i,t}^{G,out} + p_{i,t}^{PV} + p_{i,t}^{S,dch} - p_{i,t}^{S,ch} = p_{i,t}^{L,fix} + p_{i,t}^{L,flex} + p_{i,t}^{EV} \quad i \in S^N, t \in S^T \quad (4)$$

where $p_{i,t}^{PV}$ [kW] is power from photovoltaic plant, $p_{i,t}^{S,ch}$ [kW] and $p_{i,t}^{S,dch}$ [kW] are the power flows relevant to charging and discharging the storage, $p_{i,t}^{L,fix}$ [kW] is the fixed power load which cannot be controlled, $p_{i,t}^{L,flex}$ [kW] is the flexible power load portion, i.e. that load that can be shifted along the day, and $p_{i,t}^{EV}$ [kW] is the charging power of the electric vehicle.

The power exchange with the main grid is operated by

$$0 \leq p_{i,t}^{G,out} \leq \bar{P}_i^G \quad i \in S^N, t \in S^T \quad (5)$$

$$0 \leq p_{i,t}^{G,in} \leq \bar{P}_i^G \quad i \in S^N, t \in S^T \quad (6)$$

which upper bound the two terms defining $p_{i,t}^G$.

The constraints relevant to the energy storage system are:

$$x_{i,t}^S = x_{i,t-1}^S + \frac{\Delta}{CAP_i^S} \left(\Gamma_i^{S,ch} p_{i,t}^{S,ch} - \frac{1}{\Gamma_i^{S,dch}} p_{i,t}^{S,dch} \right) \quad i \in S^N, t \in S^T \quad (7)$$

$$0 \leq p_{i,t}^{S,ch} \leq \bar{P}_i^S \quad i \in S^N, t \in S^T \quad (8)$$

$$0 \leq p_{i,t}^{S,dch} \leq \bar{P}_i^S \quad i \in S^N, t \in S^T \quad (9)$$

$$\underline{x}_i^S \leq x_{i,t}^S \leq \bar{x}_i^S \quad i \in S^N, t \in S^T \quad (10)$$

where $x_{i,t}^S$ [-] is the ESS state of charge, CAP_i^S [kWh] its capacity, $\Gamma_i^{S,ch}$ and $\Gamma_i^{S,dch}$ its charging and discharging efficiencies. Constraints (8)–(10) represent upper and lower bounds of power flows and the states of charge, which are $\bar{P}_i^S$, $\bar{x}_i^S$, and $\underline{x}_i^S$, respectively. As defined by the law, the only constraint relevant to power production is that the source has to be strictly from renewable plants; the same applies to the storage. Therefore, in the specific case of ECs, the ESS usage has some limitations: for the ESS energy to count for the shared energy calculation, it must be totally from renewable. This means that the storage can be charged by withdrawing power from the main grid, but its discharge must be always and only used to satisfy the internal loads and never sold back to the to the grid. The motivation for this is that it cannot be proven with certainty what the energy mix is composed of and the energy injection that counts for the shared energy is only renewable. Nevertheless, this condition is verified as long as the buying cost from the grid is greater than the benefits (energy selling and EC incentive); indeed, in this case buying energy from the grid to charge the storage and sell the stored energy back is never convenient, but it may be convenient to charge the storage and use the stored energy later to satisfy internal loads and maximize the shared energy and the related incentive.

The constraints relevant to the deferrable loads are as follows

$$\sum_{t \in S^T} p_{i,t}^{L,flex} \Delta \geq E_i^{L,flex} \quad i \in S^N \quad (11)$$

$$0 \leq p_{i,t}^{L,flex} \leq \bar{P}_i^{L,flex} \quad i \in S^N, t \in S^T \quad (12)$$

Eq. (11) imposes that the energy required by this kind of load is delivered over the entire optimization horizon. Please note that using the greater equal is preferred over the equal to alleviate the problem; this choice is further justified if one considers that the buying price from the network is always greater than the overall benefit (selling price+EC incentive) thus the solver for sure tends to satisfy $E_i^{L,flex}$ as much as enough without exceeding. Then, the power devoted to satisfying the flexible load is subjected to bound constraints in (12).

The constraints relevant to the electric vehicle (EV) are:

$$x_{i,t}^{EV} = x_{i,t-1}^{EV} + \frac{\Delta}{CAP_i^{EV}} \Gamma_i^{EV} p_{i,t}^{EV} \quad i \in S^N, t \in S^T \quad (13)$$

$$0 \leq p_{i,t}^{EV} \leq \bar{P}_i^{EV} \quad i \in S^N, t \in S^T \quad (14)$$

$$\underline{x}_i^{EV} \leq x_{i,t}^{EV} \leq \bar{x}_i^{EV} \quad i \in S^N, t \in S^T \quad (15)$$

$$x_{i,t}^{EV} \geq x_i^{EV*} \quad i \in S^N, t = T_i^{EV*} \quad (16)$$

where $x_{i,t}^{EV}$ [-] is the EV state of charge, CAP_i^{EV} [kWh] its capacity, Γ_i^{EV} its charging efficiency. Please note that in constraint (13) only the charging mode is considered (V2G mode is discarded) as in ECs the only energy resource allowed is from RES; therefore, EVs cannot work in V2G mode because in this case there would be no guarantee that such discharging energy is fully renewable. Constraints (14) and (15) represent the upper and lower bounds of power flows and states of charge. Finally, constraint (16) ensures that the desired value for the EV state of charge is reached (and possibly exceeded) at the desired time instant T_i^{EV*} . Therefore, by knowing the initial and the final desired state of charge, an energy request is assigned to each vehicle.

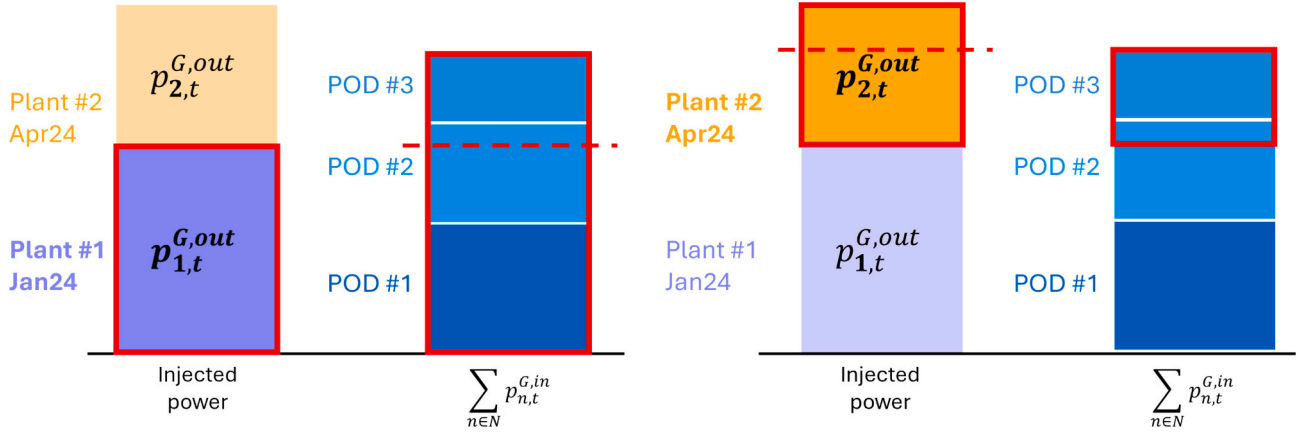


Fig. 4. Example of the comparison between the sold power and the purchased one in the proposed incentive scheme model.

4.2. High-level: ECM

The high-level of the proposed work is devoted to modelling the ECM problem, which is maximizing the overall EC benefits. To reach this goal, the energy community manager aims to maximize shared energy (1), which is indeed rewarded according to the incentive scheme described in Section 3.

Modelling the new incentive scheme is complex because the incentive depends on the plant size and therefore each plant has its own incentive value. Therefore, it is necessary to consider how much the plants that could potentially inject some power actually contribute to sharing energy, because they are incentivized on this amount of energy. To calculate the energy that each prosumer/producer contributes to share it is necessary to consider how much load has already been covered by the other prosumers/producers that count first by date of first connection to the grid. Fig. 4 provides a simple example of two production plants and three points of delivery to better understand how this incentive scheme has been modelled. As imposed by the Italian legislation, plants are ordered according to the date of the first connection to the grid. Therefore, starting from Plant #1, its production is compared with the overall withdrawn power and the minimum is taken (remember that shared energy is the minimum between electricity injected and the electricity withdrawn). Then for Plant #2, its production is not compared with the overall withdrawn power but with the residual one, i.e. is the overall one minus that already covered by Plant #1. Again, the minimum must be taken; however, in this case the production of Plant #2 exceeds the residual withdrawn power, because most of the consumption is already covered by Plant #1 which is counted first. In this way, taking the minimum, the second plant is not incentivized on its total production but on how much it actually contributes to energy sharing.

In order to model this incentive scheme, it is necessary to compare the power produced by each production plant p (ordered by their date of first connection) with the residual withdrawn energy, which is the total one minus that already covered by plants $\{1, \dots, p-1\}$. Hence, the remaining load when analyzing production plant p can be written as:

$$\sum_{n \in S^N} p_{n,t}^{G,in} - \sum_{i=1}^{p-1} p_{i,t}^{G,out} \quad (17)$$

This difference takes a negative value if the total consumption is already satisfied by the production. Since it is not possible to generate more shared energy, the maximum between the value of this difference and zero is considered.

Moreover, the result of this maximum, i.e. the load that can be virtually satisfied, must be compared with the power injected by plant p into the grid ($p_{p,t}^{G,out}$). This comparison is made by taking the minimum

between the two terms recalling the definition of shared energy in (1) and ensures that plants are not incentivized on total production but on how much they actually contribute to energy sharing.

The result is that for each hour the incentive is:

$$\sum_{p \in S^P} TIP_{p,t} \Delta \cdot \min \left\{ p_{p,t}^{G,out}, \max \left[0, \sum_{n \in S^N} p_{n,t}^{G,in} - \sum_{i=1}^{p-1} p_{i,t}^{G,out} \right] \right\} \quad (18)$$

where

- $TIP_{p,t}$ is the incentive expressed in [€/kWh], calculated for each member as in (2). Since the final value of the incentive depends on fixed and a-priori known parameters, this can be calculated in pre-processing;
- $S^P \subseteq S^N$ in the sub-set of participants with a production plant that, depending on production and consumption, could potentially inject some power;
- $p_{p,t}^{G,out(in)}$ is the power injected (withdrawn) in (from) the main grid by the p -th participant.

The community manager aims at maximizing the overall incentive obtained by the CER over the entire optimization horizon, therefore the ECM objective function would be:

$$\max J^{ECM} \quad (19)$$

with

$$J^{ECM} = \sum_{t \in S^T} \left\{ \sum_{p \in S^P} TIP_{p,t} \Delta \cdot \min \left\{ p_{p,t}^{G,out}, \max \left[0, \sum_{n \in S^N} p_{n,t}^{G,in} - \sum_{i=1}^{p-1} p_{i,t}^{G,out} \right] \right\} \right\} \quad (20)$$

4.3. Incentive scheme approximation

Since the objective function in (20) is highly nonlinear and much complicated than in our previous work [28], it cannot be implemented as it is. To cope with this issue, it has been simplified thanks to some approximation and linearization techniques. First, the inner $\max(\bullet)$ function can be substituted by introducing a variable β such as:

$$J^{ECM} = \sum_{t \in S^T} \left\{ \sum_{p \in S^P} TIP_{p,t} \Delta \cdot \min \left\{ p_{p,t}^{G,out}, \beta_{p,t} \right\} \right\} \quad (21)$$

$$\beta_{p,t} \geq \sum_{n \in S^N} p_{n,t}^{G,in} - \sum_{i=1}^{p-1} p_{i,t}^{G,out} \quad (22)$$

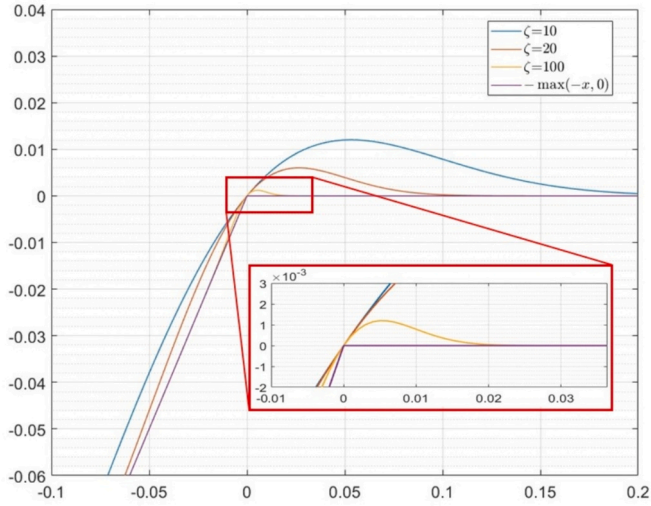


Fig. 5. Comparison of the SMU approximation by changing the parameter ζ .

$$\beta_{p,t} \geq 0 \quad (23)$$

Then, by applying the well-known rule $\min(x_1, x_2) = -\max(-x_1, -x_2)$, the min function can be rewritten as a max function obtaining:

$$J^{ECM} = \sum_{t \in S^T} \left\{ \sum_{p \in S^p} TIP_{p,t} \Delta \cdot \left[-\max \left\{ -p_{p,t}^{G,out}, -\beta_{p,t} \right\} \right] \right\} \quad (24)$$

At this point, the max function in (24) can be substituted with two techniques.

4.3.1. MILP formulation

The first one involves the absolute value as in (25). In fact, the max function can be written as

$$\max(x_1, x_2) = \frac{(x_1 + x_2) + |x_1 - x_2|}{2} \quad (25)$$

By substituting this expression in (24), the following equation is obtained

$$J^{ECM} = \sum_{t \in S^T} \left\{ \sum_{p \in S^p} TIP_{p,t} \Delta \cdot \left[-\frac{(-p_{p,t}^{G,out} - \beta_{p,t}) + |-p_{p,t}^{G,out} - \beta_{p,t}|}{2} \right] \right\} \quad (26)$$

With this technique, a MILP problem is defined.

4.3.2. NLP formulation

The second technique here proposed is the one presented in [29], where authors describe how the Gaussian error function

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt \quad (27)$$

can be used to approximate the max function such as

$$\max(x_1, x_2) = f(x_1, x_2, \zeta) = \frac{(x_1 + x_2) + (x_1 - x_2) \text{erf}(\zeta(x_1 - x_2))}{2} \quad (28)$$

By using this technique, the following objective function is obtained.

$$J^{ECM} = \sum_{t \in S^T} \left\{ \sum_{p \in S^p} TIP_{p,t} \Delta \cdot \left[-\frac{(-p_{p,t}^{G,out} - \beta_{p,t}) + (-p_{p,t}^{G,out} + \beta_{p,t}) \text{erf}(\zeta(-p_{p,t}^{G,out} - \beta_{p,t}))}{2} \right] \right\} \quad (29)$$

With this technique, a NLP problem is defined.

It must be noted that the quality of the approximation depends on the value of the parameter ζ ; in Fig. 5 a comparison of different values is reported. As clearly represented, the higher the parameter value, the better the approximation quality. For the case study, a value of $\zeta = 100$ is selected because it is sufficient for an acceptable approximation.

At a first look, someone may think that the different techniques applied to remove the $\max(\min(\max(\bullet)))$ structure (i.e. the substitution in (21), and the two formulation above) could be used in other combinations. In fact, it is possible to use them in different orders. However, due to the particular structure of the functions, by using the two techniques in 4.3.1 and 4.3.2) before using the variable β , would define a problem where the two approximations 4.3.1 and 4.3.2 are in the constraints of the problem. This solution is harder to solve and leads to numerical issues even in small instances. Instead, with the proposed order, the approximations remain in the cost function resulting in a run-time reduction.

5. Optimization problem

To consider the multi-agent framework, a multi-level approach is adopted in this paper. In this section, the problems relevant to the low-level are rewritten in the canonical form and then converted into the corresponding KKT conditions. Before that, in the following paragraph, the basic notions of multi-level optimization are reported, which will then be implemented to solve the problem.

5.1. Multi-level programming

Many decision problems involve multiple decision makers with different objectives; therefore, Decision Makers (DMs) have their own optimization goals, but share aspects of the system model and/or certain constraints. It could happen that decisions are structured hierarchically across various levels, meaning that the choices of some DMs affect the others'; this is the case when multilevel programming can be applied. The most common structure is the bilevel problem, where two hierarchical levels are considered: it involves a leader (upper level) and one or more followers (lower level), who pursue their individual goals while considering the decisions of the leader above. In this context, a potential resolution method is to integrate the different optimization problems into a single one by analytically solving the lower-level problems and embedding these solutions as constraints within the leader's problem. The general formulation of a multilevel programming problem is provided by:

$$\begin{aligned} & \underset{x}{\text{Min}} f(x, x_1, \dots, x_N) \\ & \text{s.t.} \\ & g(x, x_1, \dots, x_N) = 0 \\ & h(x, x_1, \dots, x_N) \leq 0 \end{aligned} \quad (30)$$

$$\left\{ \begin{array}{l} \underset{x_1}{\text{Min}} f_1(x, x_1, \dots, x_N) \\ \text{s.t.} \\ g_1(x, x_1, \dots, x_N) = 0 \\ h_1(x, x_1, \dots, x_N) \leq 0 \end{array} \right. \dots \left\{ \begin{array}{l} \underset{x_N}{\text{Min}} f_N(x, x_1, \dots, x_N) \\ \text{s.t.} \\ g_N(x, x_1, \dots, x_N) = 0 \\ h_N(x, x_1, \dots, x_N) \leq 0 \end{array} \right. \quad (31)$$

where vector $x \in \mathbb{R}^n$ includes the set of optimization variables of the upper level problem, vector $x_i \in \mathbb{R}^m$ includes the set of optimization variables of constraining problem i , and $\lambda_i \in \mathbb{R}^m$ and $\mu_i \in \mathbb{R}^m$ are the dual variable vectors of problem i associated with the equality and inequality constraints. The analytical solution of each follower' optimization problem can be found through KKT conditions:

$$\begin{aligned}
& \underset{x}{\text{Min}} f(x, x_1, \dots, x_N) \\
& \text{s.t.} \\
& g(x, x_1, \dots, x_N) = 0 \\
& h(x, x_1, \dots, x_N) \leq 0 \\
& \text{KKT}_1 \\
& \vdots \\
& \text{KKT}_N
\end{aligned} \quad (32)$$

It is reminded that the KKT conditions of a general problem (33):

$$\begin{aligned}
& \underset{x}{\text{min}}_x f(x) \\
& \text{s.t.} \\
& h(x) = 0 \\
& g(x) \leq 0
\end{aligned} \quad (33)$$

are

$$\begin{aligned}
& \nabla_x f(x) + \lambda^T \nabla_x h(x) + \mu^T \nabla_x g(x) = 0 \\
& h(x) = 0 \\
& g(x) \leq 0 \\
& \mu^T g(x) = 0 \\
& \mu \geq 0
\end{aligned} \quad (34)$$

The first KKT condition, known as the stationarity condition, ensures that the derivative of the objective function with respect to any admissible direction, considering the constraints, must be zero i.e. no local improvement is possible. The second and third KKT conditions are related to primal feasibility: both the equality and inequality constraints must be satisfied for the optimal solution to lie within the feasible region. The fourth condition is the complementary slackness condition, ensuring that if the constraint is active the associated multiplier μ is positive, or if the constraint is inactive $\mu = 0$. The last condition, known as dual feasibility condition, prevents negative Lagrange multipliers associated with the inequality constraints; this would suggest that violating a constraint improves the solution, which contradicts the very definition of constraints.

5.2. The overall optimization problem

With the notions just reviewed, the overall problem is given by the high-level objective function and the low-level problems converted into the corresponding KKT. To do this, the ECPs problems must be reformulated, as shown in this paragraph.

Since the problem statement is the same for each participant, for the sake of simplicity, the index relevant to each ECP is omitted in this section. Moreover, the bold characters in the following will denote vectors and matrices.

By substituting the power balance (4) in (3) and rearranging, the objective function becomes

$$\begin{aligned}
J^{ECP} = & \Delta \left[(\mathbf{C}^{\text{buy}} - \mathbf{C}^{\text{sell}})^T \mathbf{p}^{\text{G,out}} + (\mathbf{C}^{\text{buy}})^T (\mathbf{p}^{\text{S,ch}} - \mathbf{p}^{\text{S,dch}} + \mathbf{p}^{\text{L,flex}} + \mathbf{p}^{\text{EV}}) \right] \\
& + \alpha \cdot (\mathbf{x}_T^{\text{S}} - \mathbf{x}_0^{\text{S}})^2
\end{aligned} \quad (35)$$

where the five variables represent the control variables such as

$$\mathbf{u} = \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \mathbf{u}_3 \\ \mathbf{u}_4 \\ \mathbf{u}_6 \end{bmatrix} = \begin{bmatrix} \mathbf{p}^{\text{G,out}} \\ \mathbf{p}^{\text{S,ch}} \\ \mathbf{p}^{\text{S,dch}} \\ \mathbf{p}^{\text{L,flex}} \\ \mathbf{p}^{\text{EV}} \end{bmatrix} \quad (36)$$

Then, (35) can be written as a function of $\mathbf{u}$

$$\begin{aligned}
J^{ECP} = & \Delta \left[(\mathbf{C}^{\text{buy}} - \mathbf{C}^{\text{sell}})^T \quad (\mathbf{C}^{\text{buy}})^T \quad (-\mathbf{C}^{\text{buy}})^T \quad (\mathbf{C}^{\text{buy}})^T \quad (\mathbf{C}^{\text{buy}})^T \right] \mathbf{u} \\
& + \alpha \cdot \boldsymbol{\Theta}^T \boldsymbol{\Theta} \mathbf{u}
\end{aligned} \quad (37)$$

where:

$$\boldsymbol{\Theta} = \begin{bmatrix} \mathbf{0} & \mathbf{K}^{\text{ch}} & \mathbf{K}^{\text{dch}} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (38)$$

Equality constraints are rewritten in matrix form to be substituted in the inequality constraints, so that the overall optimization problem becomes an *inequality constrained convex optimization problem*. The power balance in (4) could be written as

$$[\mathbf{I} \quad \mathbf{I} \quad -\mathbf{I} \quad \mathbf{I} \quad \mathbf{I}] \mathbf{u} = \mathbf{P}^{\text{PV}} - \mathbf{P}^{\text{L,fix}} + \mathbf{p}^{\text{G,in}} \quad (39)$$

Then, the ESS and EV state equations in (7) and (13) could be written as

$$\mathbf{x}^{\text{S}} = \begin{bmatrix} \mathbf{0} & \mathbf{K}^{\text{ch}} & \mathbf{K}^{\text{dch}} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{u} + \mathbf{1} \mathbf{X}_0^{\text{S}} \quad (40)$$

$$\mathbf{x}^{\text{EV}} = \begin{bmatrix} \mathbf{0} & \mathbf{K}^{\text{ch}} & \mathbf{K}^{\text{dch}} & \mathbf{0} & \mathbf{0} \end{bmatrix} \mathbf{u} + \mathbf{1} \mathbf{X}_0^{\text{EV}} \quad (41)$$

where terms $\mathbf{K}^*$ are TxT matrixes defined as:

$$\mathbf{K}^{\text{ch}} = \begin{cases} k_{ij} = 0 & \text{if } i < j \\ k_{ij} = \frac{\Gamma^{\text{S,ch}} \Delta}{\text{CAP}^{\text{S}}} & \text{if } i \geq j \end{cases} \quad (42)$$

$$\mathbf{K}^{\text{dch}} = \begin{cases} k_{ij} = 0 & \text{if } i < j \\ k_{ij} = -\frac{\Delta}{\Gamma^{\text{S,dch}} \text{CAP}^{\text{S}}} & \text{if } i \geq j \end{cases}$$

$$\mathbf{K}^{\text{EV}} = \begin{cases} k_{ij} = 0 & \text{if } i < j \\ k_{ij} = \frac{\Gamma^{\text{EV}} \Delta}{\text{CAP}^{\text{EV}}} & \text{if } i \geq j \end{cases} \quad (43)$$

Note that the formulation in (40)–(43) represents the conversion of the dynamic problem to a static one. Then, making appropriate substitutions and some algebraic manipulations, inequality constraints (5), (6), (8)–(12), (14)–(16) can be written in matrix form such as

$$\boldsymbol{\Omega} \mathbf{u} + \boldsymbol{\Psi} \leq \mathbf{0} \quad (44)$$

with

$$\boldsymbol{\Omega} = \begin{bmatrix} \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{K}^{\text{ch}} & -\mathbf{K}^{\text{dch}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}^{\text{ch}} & \mathbf{K}^{\text{dch}} & \mathbf{0} & \mathbf{0} \\ -\mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -\mathbf{I} & -\mathbf{I} & \mathbf{I} & -\mathbf{I} & -\mathbf{I} \\ \mathbf{I} & \mathbf{I} & -\mathbf{I} & \mathbf{I} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{1} \Delta & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{K}^{\text{EV}} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{K}^{\text{EV}} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{K}_{\text{PEV}}^{\text{EV}} \end{bmatrix} \quad \boldsymbol{\Psi} = \begin{bmatrix} \mathbf{0} \\ -\bar{\mathbf{P}}^{\text{S}} \\ \mathbf{0} \\ -\bar{\mathbf{P}}^{\text{S}} \\ \mathbf{x}^{\text{S}} - \mathbf{1} \mathbf{X}_0^{\text{S}} \\ -\bar{\mathbf{x}}^{\text{S}} + \mathbf{1} \mathbf{X}_0^{\text{S}} \\ \mathbf{0} \\ -\bar{\mathbf{P}}^{\text{G}} \\ \mathbf{p}^{\text{PV}} - \mathbf{p}^{\text{L,fix}} \\ -\bar{\mathbf{P}}^{\text{G}} - \mathbf{p}^{\text{PV}} + \mathbf{p}^{\text{L,fix}} \\ \mathbf{0} \\ -\bar{\mathbf{P}}^{\text{L,flex}} \\ \mathbf{E}^{\text{L,flex}} \\ \mathbf{0} \\ -\bar{\mathbf{P}}^{\text{EV}} \\ \mathbf{x}^{\text{EV}} - \mathbf{1} \mathbf{X}_0^{\text{EV}} \\ \mathbf{1} \mathbf{X}_0^{\text{EV}} - \bar{\mathbf{x}}^{\text{EV}} \\ \mathbf{x}^{\text{EV}+} \end{bmatrix}$$

where double inequalities have been split to present the constraints explicitly. Again, please note that the low-level equality constraints do not explicitly enter the definition of these matrixes as they are substituted in the objective function and in other constraints.

Table IV
General data

T	Δ [h]	$\bar{P}^{flex}$ [kW]	$\bar{P}^G$ [kW]
24	1	2	30
Γ^{sch} [/]	Γ^{sch} [/]	$\bar{X}^S$ [/]	$\bar{X}^S$ [/]
0.95	0.95	0.2	1
$\bar{P}^{EV}$ [kW]	Γ^{EV} [/]	$\bar{X}^{EV}$ [/]	$\bar{X}^{EV}$ [/]
3	0.95	0.2	1

5.3. Deriving KKT conditions

Once the ECPs problems are properly reformulated, it is possible to define the KKT conditions as

$$\begin{bmatrix} (\mathbf{C}^{buy} - \mathbf{C}^{sell})^T & (\mathbf{C}^{buy})^T & (-\mathbf{C}^{buy})^T & (\mathbf{C}^{buy})^T & (\mathbf{C}^{buy})^T \end{bmatrix} \Delta + \mu^T \Omega = 0 \quad (45)$$

$$\mu^T \{\Omega \mathbf{u} + \Psi\} = 0 \quad (46)$$

$$\mu \geq 0 \quad (47)$$

where μ is the Lagrange multiplier associated with inequality constraints. Of course, also (44) must be considered among KKT conditions.

Note that, to obtain a MILP problem, (46) can be linearized by introducing a binary variable δ and substituting the constraints with

$$\{\Omega \mathbf{u} + \Psi\} + (1 - \delta) \cdot M \geq 0 \quad (48)$$

$$\mu - \delta M \leq 0 \quad (49)$$

where M is a sufficiently big number.

Finally, depending on the considered technique (4.3.1 or 4.3.2) the overall optimization problem becomes:

- MILP formulation: (19), (26) s.t. (22)–(23), (44)–(45), (47)–(49).
- NLP formulation: (19), (29) s.t. (22)–(23), (44)–(47).

6. Application to a case study

This section presents the case study and discusses the main results. Then, the comparison between the two approximation techniques introduced in paragraphs 4.3.1 and 4.3.2 is presented in order to show their performances. A scalability analysis is also presented to prove the suitability of this approach for larger scenarios. Finally, in the last part of this section, a sensitivity analysis on the most important parameters is described.

6.1. Case study application

This paragraph presents the main data relevant to the analyzed EC; in particular, a medium-size case study is tested, where an EC with 10 ECPs

Table V
ECP data

	CAP^S [kWh]	X_0^S [/]	$\bar{P}^S$ [kW]	E^{flex} [kWh]	CAP^{EV} [kWh]	X_0^{EV} [/]	X^{EV*} [/]	T^{EV*} [h]
ECP1	/	/	/	3.21	10	0.4	0.8	8
ECP2	30	0.2	10	7.01	10	0.5	0.8	11
ECP3	20	0.2	10	11.31	/	/	/	/
ECP4	/	/	/	8.31	10	0.2	0.8	16
ECP5	/	/	/	6.56	/	/	/	/
ECP6	15	0.2	10	5.14	10	0.3	0.9	21
ECP7	/	/	/	3.20	/	/	/	/
ECP8	/	/	/	12.76	/	/	/	/
ECP9	/	/	/	6.50	10	0.35	0.8	13
ECP10	/	/	/	13.78	/	/	/	/

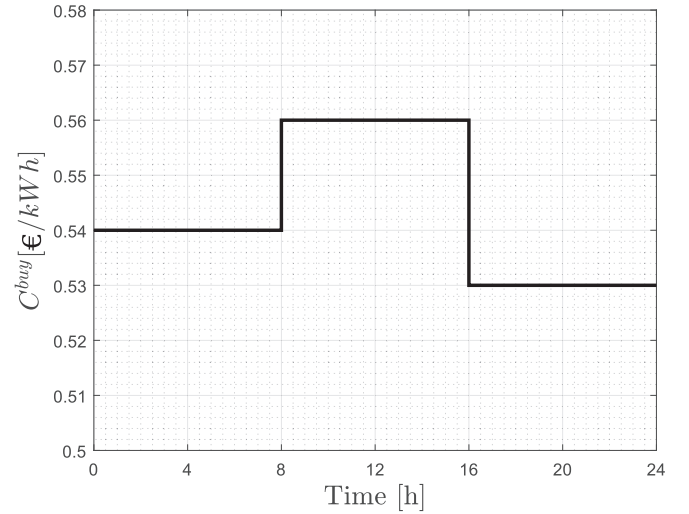
($N = 10$) is considered. Table IV reports the general data relevant to the optimization problem, while Table V shows the data relevant to each ECP. Please note that selecting a one-hour time step is coherent with the legislation, according to which shared energy is calculated ex-post on an hourly basis.

In addition to these data, it is necessary to specify the selling and buying price for exchanging power with the main grid: C_t^{sell} is set to 0.08 [€/kWh], while C_t^{buy} is reported in Fig. 6.

Moreover, Fig. 7 depicts the power profiles relevant to the PV production and the fixed load for each ECP.

By analyzing data in Table V and Fig. 7, one can notice how generation and consumption are allocated: as an example, only ECP 1–2–4–6–9 need to charge their EVs, while ECP 2–3–6 are the only ones to have a production plant and so they are the ones that could potentially inject some power into the grid.

The incentive value must also be specified, referring to the equation provided by the legislation as in (2). In this paper, all the production plants are assumed to be smaller than 200 kW, which is consistent with small PV plants that are installed on the roofs of independent homes and condominiums. Therefore, according to Table II, $CAP_i = 0.1$ [€/kWh] and $TP_i^{base} = 0.08$ [€/kWh] for all plants. As far as the correction factor is concerned, it is assumed that the considered EC is in the north of Italy and so $FC^{zone} = 0.010$ [€/kWh]. Then, since the disbursement of the possible payout of a capital contribution does not affect the optimal solution, parameter F is assigned a value equal to zero. Finally, considering the values just defined for the parameters in eq. (2) and the energy price in Fig. 6, the final value for the incentive TIP_{it} is 0.11 [€/kWh] for all plants and in every time interval.

**Fig. 6.** Energy buying price.

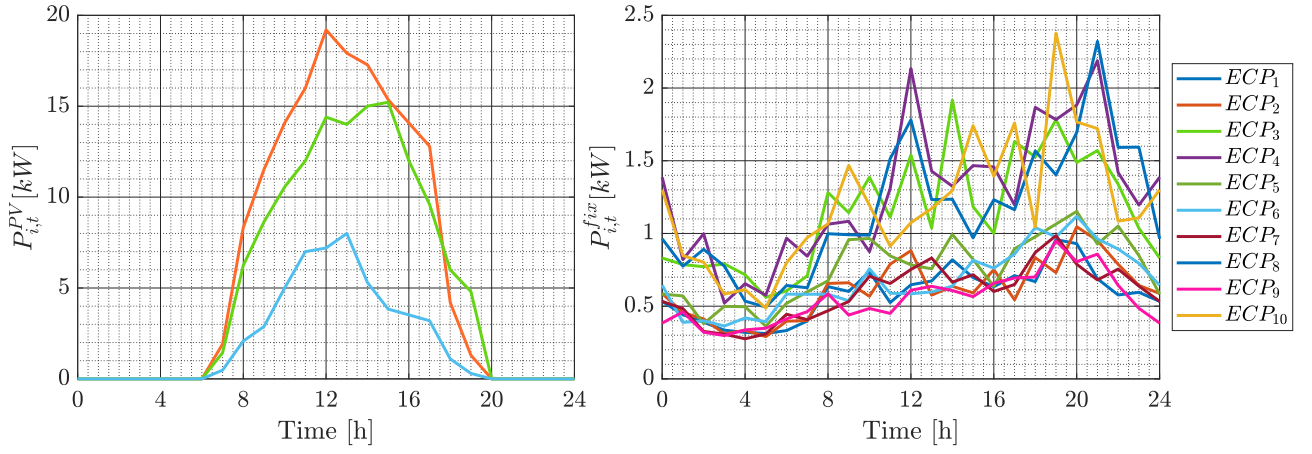


Fig. 7. ECP profiles on PV production and consumption.

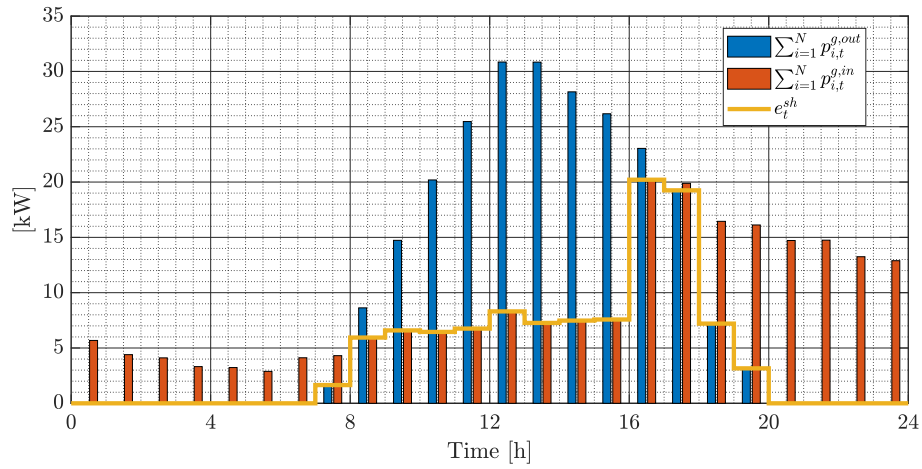


Fig. 8. Power exchange with the grid and resulting shared energy.

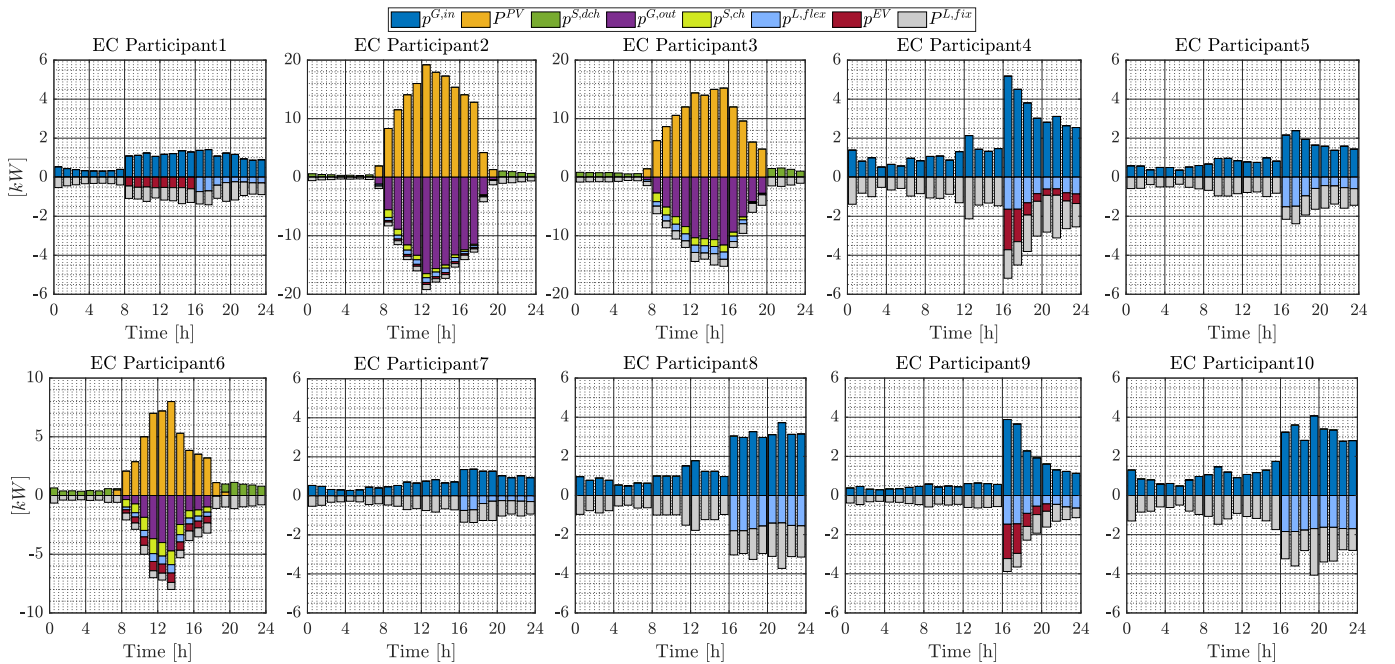


Fig. 9. Power balance of each EC Participant.

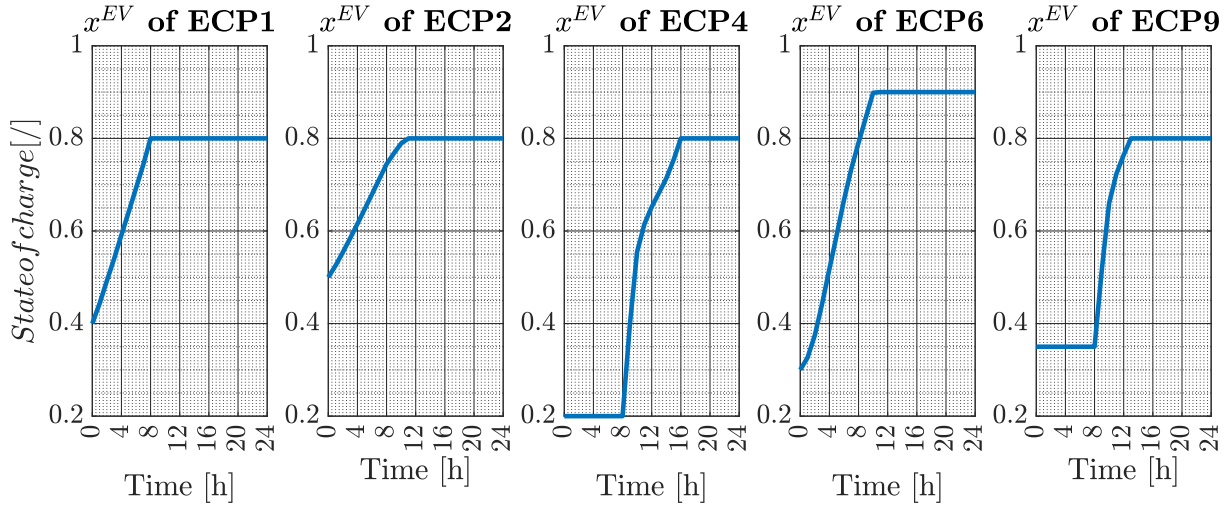


Fig. 10. EV state of charge for each EC Participant.

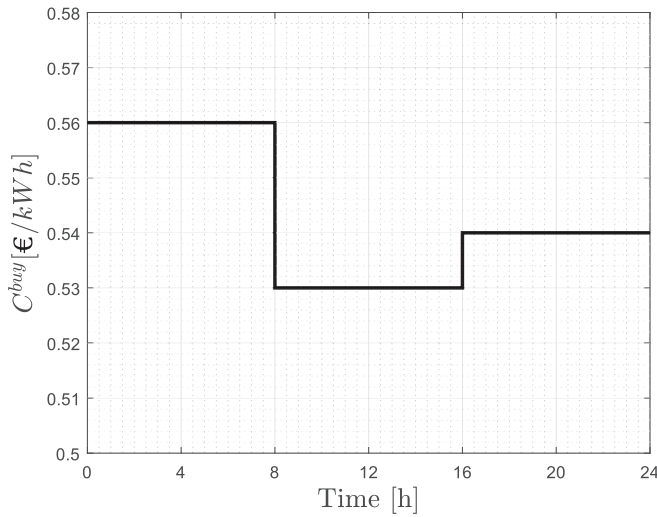


Fig. 11. Modified energy buying price.

6.2. Optimization results

In this paragraph, the proposed methodology is tested on the instance just described and a detailed description of the main results is provided. It is relevant to highlight that, the results illustrated below are obtained by applying the second approximation approach 4.3.2 and therefore are relevant to a NLP problem. This approximation method is preferred over the one leading to a MILP formulation (4.3.1) because it performs better, as will be explained in a dedicated paragraph 6.3.

By applying the second approximation approach in 4.3.2, the resulting problem is an NLP and the solution is obtained via IPOPT solver within the YALMIP toolbox [30] in MATLAB on an AMD Ryzen 75800H, 3.2 GHz, 8 core processor, 16GB RAM in about 28 s. In Table VI the total cost for each user is reported.

Fig. 8 reports the amount of shared energy in each hour. As presented in eq. (1), this quantity (yellow line the figure) is always equal to the minimum between the overall power absorption (red bars) and the overall power injection (blue bars). Please note that energy and power coincide in this specific case, where the selected Δ is one hour. This figure allows also to appreciate that in the extreme hours of the day, when there is no production from PV (remember that, by law, RES generation is the only one allowed), the shared energy is null as no EPC

is selling power to the grid. This means that for the shared energy to be non-zero, there must be simultaneous consumption and production.

The total shared energy corresponds to 107.84 kWh, resulting in an overall incentive (for the whole EC) of 11.86 €. This incentive could seem not very high but, considering the costs in Table VI, it is possible to see that the overall expenses for the EC members is 97.71 € and thus the incentive corresponds to a 12.14 % reduction in the total electricity bill. Moreover, note that the actual benefit for each ECP depends on the splitting rules of each REC.

Fig. 9 reports the power balance for each ECP. In addition to verifying that the balance is respected at any time step, this figure allows also to verify how the ESS is exploited for those participants who have it: the storage is charged during the central hours of the day (excess of PV production) and discharged to satisfy the load. This is the usual operation of ESS in generic households since this approach can reduce the costs of the owner. This figure also allows to see that the storage is only charged by renewable energy. This behavior complies with the regulation of RECs even if the use of the storage does not provide any contribution to the shared energy since the stored energy is used only to satisfy the load of the same user; indeed, this represents a more convenient solution in terms of operation costs.

In Fig. 10, the EV charging profiles are reported. The figure, allows to verify that constraint (16) is respected: the desired state of charge is reached within the predefined time instant T^{EV*} . Of course, the plotted profiles always increase since for EVs vehicle-to-grid mode is discarded to comply with the request for only renewable energy production.

Another interesting analysis is relevant to the impact of the shared energy incentive on the optimal energy management for the users, i.e. how the ECP scheduling changes whether the objective of shared energy maximization (20) is considered or not. In fact, it should be considered that even in the case where the objective function (20) is not considered and the only cost to be minimized is that of the members, there is still a certain amount of shared energy provided by the optimal scheduling of ECPs. In particular, when (20) is disregarded the shared energy amounts to 96.5 kWh. As already presented, by introducing the shared energy maximization it is possible to obtain an increase of 10.17 kWh (+10.53 %) without changing the cost for the ECPs.

In order to provide more insights about the behavior of the system, in particular about the impact of the storage system, a comparison between the presented scenario and a situation where the ECPs do not have any storage is presented in Table VII.

The huge advantage brought from the storage (−11.15 % of the total ECPs' costs) is particularly evident within this system configuration due to the nature of the considered renewable plants which can benefit from the flexibility induced by the storage. On the shared energy side, it is

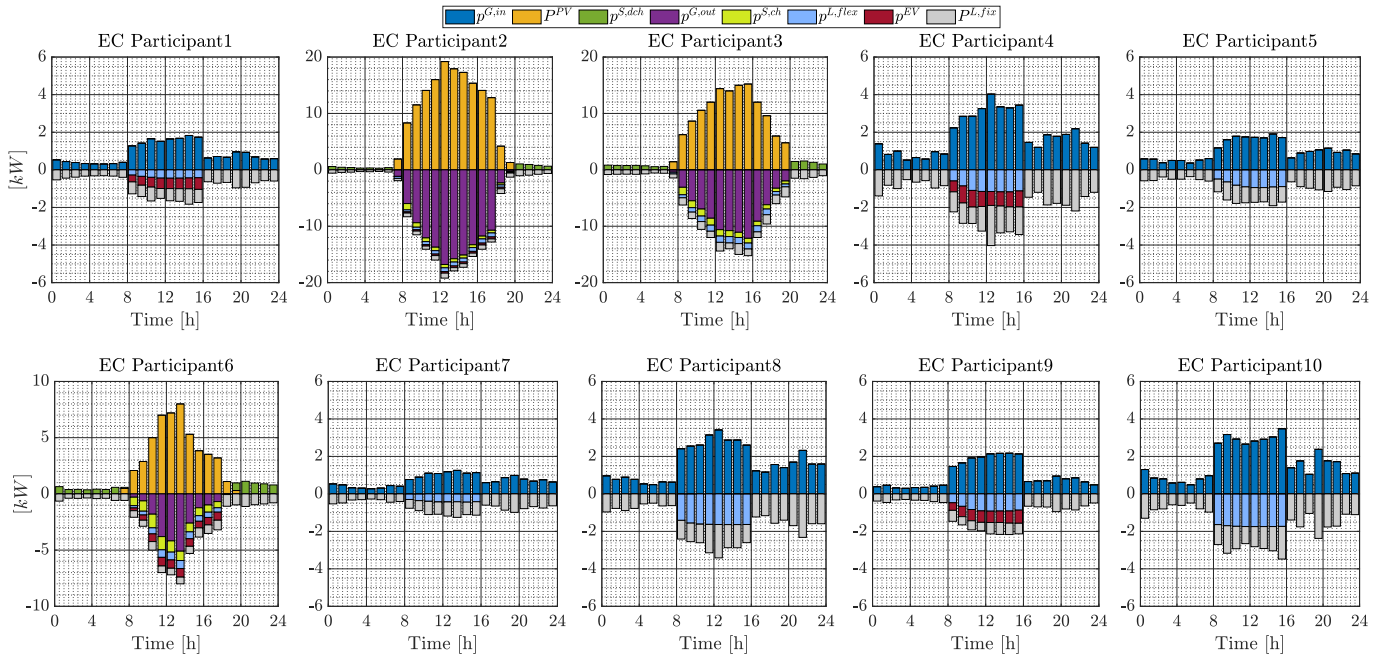


Fig. 12. ECPs' power balance in a modified multi-rate energy pricing framework.

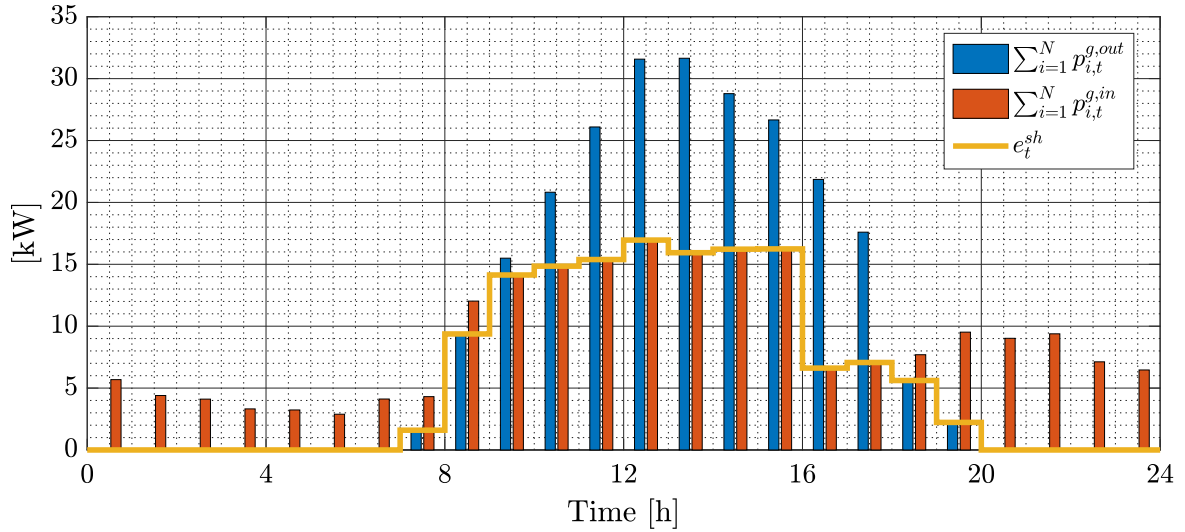


Fig. 13. Power exchange with the grid and resulting shared energy.

possible to find that without the ESS the total value increases from 107.84 kWh to 115.20 kWh, corresponding to an increase of 0.75 € (+6.8 %) in the received incentive. This behavior confirms that the savings due to the avoided purchase of energy influence the solution more than the incentives relevant to the shared energy.

6.3. Comparison of approximation techniques and scalability analysis

This section tests the performance of the two proposed approximation techniques. Since they lead to two different classes of problems, two different solvers are used: GUROBI for the approximation in Section 4.3.1 (MILP) and IPOPT for the one in Section 4.3.2 (NLP).

The two analyzed techniques are compared by running the proposed model considering problems of increasing size. From the results reported in Table VIII, it is clear that the second method, using the Gaussian error function, has the best performances: the result for the optimization problem is about the same, but it performs better from a runtime point of

view.

An interesting result of this comparison is that the proposed model can be solved quickly even for complex systems with many ECPs. Even if the existing ECs are very small nowadays, they are expected to grow in the next years, leading to the development of very large and complex systems with the need of a very fast optimization, able to run online to provide an optimal management of the whole REC. The proposed model can be a valid candidate to play this role.

6.4. Sensitivity analysis

In this section, a sensitivity analysis on two of the most impacting parameters is conducted. First, the results obtained by different electricity prices are considered. Then, the role of renewable production is investigated.

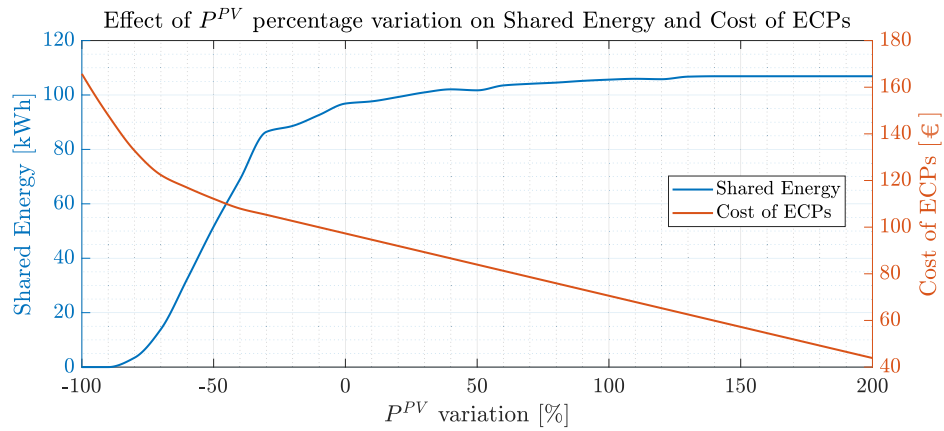


Fig. 14. Effect of the variation of renewable power on shared energy and users' cost.

Table VI

ECP Cost [€]

ECP1	ECP2	ECP3	ECP4	ECP5	ECP6	ECP7	ECP8	ECP9	ECP10	$\sum ECPs$
11.75	-10.26	-7.18	24.27	13.45	-1.71	9.51	21.94	13.11	22.82	97.71

Table VII

ECP Costs comparison with and without the ESS [€]

	ECP1	ECP2	ECP3	ECP4	ECP5	ECP6	ECP7	ECP8	ECP9	ECP10	$\sum ECPs$
ESS	11.75	-10.26	-7.18	24.27	13.45	-1.71	9.51	21.94	13.11	22.82	97.71
No ESS	11.75	-7.49	-2.49	24.27	13.45	1.74	9.51	21.94	13.11	22.82	108.61

Table VIII

Comparison of the two analyzed approximation techniques

	T = 24 N = 3		T = 24 N = 6		T = 24 N = 10		T = 24 N = 15		T = 24 N = 20		T = 24 N = 30		T = 24 N = 50	
	MILP	NLP	MILP	NLP	MILP	NLP	MILP	NLP	MILP	NLP	MILP	NLP	MILP	NLP
Solver time [s]	21,17	4,58	285,31	12,32	>1000	27,99	>1000	79,81	>1000	79,75	>1000	175,52	>1000	699,7
Obj	28,27	28,27	28,64	28,64		28,64		31,03		31,03		59,04		89,44
Err %		-5.69E-05		-1.08E-04										

6.4.1. Energy buying price

As energy buying price is probably the most important parameter, especially from the ECPs viewpoint. To analyze its impact the price already presented in Fig. 6 is modified such as the lower tier is the one corresponding to the central hours of the day, i.e. those where PV generation is concentrated (Fig. 11).

Of course, this scenario represents a more favorable situation for EC operations in terms of shared energy, in fact the best hours for purchasing electricity are also those where the higher value for renewable power production is reached. Fig. 12 presents the resulting power balance.

As expected, the power consumption is concentrated in the central hours, as a consequence the values for shared energy are higher with respect to the initial case. The behavior of the ESSs remains more or less the same. In the following, Fig. 13 illustrates the overall power exchange and the resulting shared energy calculation.

In this case, both cost term and shared energy improve. The total costs for the ECPs decrease slight from 97.71€ to 97.28€ (−0.4 %), while the total shared energy increase from 107.84 kWh to 142.22 kWh (+31.9 %). This significant increase corresponds to an increase in the received incentive of 3.5€.

This analysis clearly presents how a favorable cost structure can improve the results and then the effectiveness of ECs.

6.4.2. Renewable power variation

Another critical parameter relevant to RECs can be identified as renewable production. Indeed, since shared energy is calculated based solely on renewable power production, the amount of power generated by renewables plants considerably impacts the results. To analyze the consequences of power production the forecasted power production is decreased and increased by different percentages, as illustrated in Fig. 14. Note that in this analysis the installation costs and the physical limits for the PV plants are not considered since the goal is to analyze the trend in the shared energy and the ECPs costs.

On the left side it is possible to see how the progressive reduction of renewable power leads to an increase in the overall costs and to a total absence of shared energy. On the right side, while increasing the power production, the costs decrease while the total shared energy reaches a plateau that is due to the total satisfaction of the load. In that case, even increasing production, the value of the shared energy cannot increase due to the actual definition given by the law.

7. Conclusions

This paper presents a novel approach for the optimal management of an EC, exploiting a bi-level architecture in its proposed model. At the lower level, each ECP solves an individual optimization problem aimed at minimizing its own costs. These low-level problems are integrated, via

the KKT conditions, into the high-level problem, representing the ECM whose objective is to maximize shared energy. To the best of the authors' knowledge, this work is the first to accurately model the complex cost structures introduced by recent regulatory updates. Given the complexity of the cost function, two alternative approximations are proposed.

The model is tested with a non-linear approximation on a scenario involving ten ECPs equipped with various energy technologies. The results indicate that the incentive for shared energy significantly reduces the total energy costs for community members, achieving approximately a 12 % reduction in their energy bills. While storage systems have been shown to be effective in reducing costs, their impact on shared energy remains marginal in this instance. Then, the model's scalability is analyzed and tested. This is done by employing the two proposed approximations. The results demonstrate that the non-linear approximation provides the best performance in terms of accuracy and computational feasibility. Finally, a sensitivity analysis is presented. This analysis examines the impacts of different energy pricing and renewable energy production. A key finding is that, when energy prices are lower in the central hours of the day, shared energy generation can be enhanced within the EC and consequently the establishment of new ECs is fostered. Regarding the renewable plants, the results show how an imbalance in terms of renewable production and power consumption is not beneficial to the EC operations.

The model assumes perfect knowledge of energy demand and renewable energy production. Future research could incorporate forecast uncertainty to improve its real-world applicability. Another promising avenue for future work is quantifying the broader impact of local consumption promoted by ECs on the distribution network.

CRedit authorship contribution statement

Virginia Casella: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Giulio Ferro:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Luca Parodi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Michela Robba:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work conducted for this paper was partially supported by PRIN 2022 ECODREAM (Energy Community management: DistRibuTEd Algorithms and toolboxes for efficient and sustainable operations), and DUT OPEN4CEC (Service-oriented Open Platform for Citizen Energy Communities (CEC) – a collaborative platform).

Data availability

The authors do not have permission to share data.

References

- [1] European Parliament. Renewable energy directive (RED II) on the promotion of the use of energy from renewable sources. 2018.

- [2] Valderrama DF, Ferro G, Parodi L, Robba M. A multilevel optimization model for a distribution power grid with the active participation of electric vehicles via aggregators. In: In 2023 9th international conference on control, decision and information technologies (CoDIT). IEEE; 2023. p. 1984–9.
- [3] Ahmed S, Ali A, D'Angola A. A review of renewable energy communities: concepts, scope, Progress, challenges, and recommendations. Sustainability 2024;16(5): 1749. <https://doi.org/10.3390/su16051749>.
- [4] Vecchi F, Stasi R, Berardi U. Modelling tools for the assessment of renewable energy communities. Energy Rep 2024;11:3941–62. <https://doi.org/10.1016/j.egy.2024.03.048>.
- [5] Laurini B, Bonvini B, Bracco S. Optimal design model for a public-private renewable energy community in a small Italian municipality. Sustainable Energy, Grids Networks 2024;40:101545.
- [6] Magni GU, Battistelli F, Trovati F, Groppi D, Astiaso Garcia D. How national policies influence energy community development across Europe? A review on societal, technical, and economical factors. Energy Convers Manage: X 2024;23: 100624. <https://doi.org/10.1016/j.ecmx.2024.100624>.
- [7] Lazzari F, Mor G, Cipriano J, Solsona F, Chemisana D, Guericke D. Optimizing planning and operation of renewable energy communities with genetic algorithms. Appl Energy 2023;338:120906. <https://doi.org/10.1016/j.apenergy.2023.120906>.
- [8] Cutore E, Volpe R, Sgroi R, Fichera A. Energy management and sustainability assessment of renewable energy communities: the Italian context. Energy Convers Manag 2023;278:116713. <https://doi.org/10.1016/j.enconman.2023.116713>.
- [9] Cielo A, Margiaria P, Lazzaroni P, Mariuzzo I, Repetto M. Renewable energy communities business models under the 2020 Italian regulation. J Clean Prod 2021;316:128217. <https://doi.org/10.1016/j.jclepro.2021.128217>.
- [10] Sousa J, et al. Renewable energy communities optimal design supported by an optimization model for investment in PV/wind capacity and renewable electricity sharing. Energy 2023;283:128464. <https://doi.org/10.1016/j.energy.2023.128464>.
- [11] Faria J, Marques C, Pombo J, Mariano S, Calado MDR. Optimal sizing of renewable energy communities: a multiple swarms multi-objective particle swarm optimization approach. Energies 2023;16(21):7227. <https://doi.org/10.3390/en16217227>.
- [12] Elomari Y, Mateu C, Marín-Genescà M, Boer D. A data-driven framework for designing a renewable energy community based on the integration of machine learning model with life cycle assessment and life cycle cost parameters. Appl Energy 2024;358:122619. <https://doi.org/10.1016/j.apenergy.2024.122619>.
- [13] Gruber L, Kockar I, Wogrin S. Towards resilient energy communities: evaluating the impact of economic and technical optimization. Int J Electr Power Energy Syst 2024;155:109592. <https://doi.org/10.1016/j.ijepes.2023.109592>.
- [14] Stentati M, Paoletti S, Vicino A. Optimization of energy communities in the Italian incentive system. In: In 2022 IEEE PES innovative smart grid technologies conference Europe (ISGT-Europe). Novi Sad, Serbia: IEEE; Oct. 2022. p. 1–5. <https://doi.org/10.1109/ISGT-Europe54678.2022.9960513>.
- [15] Zanvetto GG, Casini M, Giannitrapani A, Paoletti S, Vicino A. Optimal Management of Energy Communities Hosting a Fleet of electric vehicles. Energies 2022;15(22):8697. <https://doi.org/10.3390/en15228697>.
- [16] Bianchi FR, et al. Modelling and optimal management of renewable energy communities using reversible solid oxide cells. Appl Energy 2023;334:120657. <https://doi.org/10.1016/j.apenergy.2023.120657>.
- [17] Talluri G, Lozito GM, Grasso F, Iturrino Garcia C, Luchetta A. Optimal battery energy storage system scheduling within renewable energy communities. Energies 2021;14(24):8480. <https://doi.org/10.3390/en14248480>.
- [18] Conte F, D'Antoni F, Natrella G, Merone M. A new hybrid AI optimal management method for renewable energy communities. Energy and AI 2022;10:100197. <https://doi.org/10.1016/j.egyai.2022.100197>.
- [19] Faraji J, De Grève Z, Vallée F. Hierarchical energy sharing Management for a Renewable Energy Community with heterogeneous end-users. In: in 2023 IEEE Belgrade PowerTech. Belgrade, Serbia: IEEE; 2023. p. 1–8. <https://doi.org/10.1109/PowerTech55446.2023.10202668>.
- [20] Gomes L, Vale Z. Costless renewable energy distribution model based on cooperative game theory for energy communities considering its members' active contributions. Sustain Cities Soc 2024;101:105060. <https://doi.org/10.1016/j.scs.2023.105060>.
- [21] Barone G, Buonomano A, Forzano C, Palombo A, Russo G. The role of energy communities in electricity grid balancing: a flexible tool for smart grid power distribution optimization. Renew Sust Energ Rev 2023;187:113742. <https://doi.org/10.1016/j.rser.2023.113742>.
- [22] Messilem MAM, Deplano D, Franceschelli M, Usai E, Carli R. Distributed optimization for networks of battery energy storage Systems in Energy Communities with shared energy incentives. In: In 2024 IEEE 20th international conference on automation science and engineering (CASE). Bari, Italy: IEEE; Aug. 2024. p. 2745–51. <https://doi.org/10.1109/CASE59546.2024.10711732>.
- [23] Moré JJ, Deplano D, Pilloni A, Pisano A, Franceschelli M. Online coordination of BESS and Thermostatically control loads for shared energy optimization in energy communities. In: In 2024 IEEE 20th international conference on automation science and engineering (CASE). Bari, Italy: IEEE; Aug. 2024. p. 2738–44. <https://doi.org/10.1109/CASE59546.2024.10711392>.
- [24] NextGenerationEU, GSE Gestore Servizi Elettrici. Regole operative per l'accesso al servizio per l'autoconsumo diffuso e al contributo PNRR. In: MASE - Ministero dell'Ambiente e della Sicurezza Energetica; 2024 [Online]. Available: <http://www.mase.gov.it/sites/default/files/ALLEGATO%201%20Regole%20operative%20CACER%20def.pdf>.
- [25] GSE Gestore Servizi Energetici. GSE Website [Online]. Available: <https://www.gse.it/servizi-per-te/autoconsumo>.

- [26] GSE Gestore Servizi Energetici. Regole tecniche sistemi di accumulo [Online]. Available: <https://www.gse.it/servizi-per-te/fotovoltaico/sistemi-di-accumulo/documenti>; 2021.
- [27] GSE - Gestore Servizi Elettrici. Comunità Energetiche e Autoconsumo 2024. Articolo KB0016870 2024 [Online]. Available: https://supportogse.service-now.com/csm/it/kb0016870?id=faq&sys_id=0f268b8ec34246143fd7bffb050131cf.
- [28] Casella V, Ferro G, Parodi L, Robba M. Energy community optimal management: A Bilevel approach. In: In *2024 IEEE 20th international conference on automation science and engineering (CASE)*. IEEE; 2024. p. 2720–5.
- [29] Biswas K, Kumar S, Banerjee S, Pandey AK. Smooth maximum unit: Smooth activation function for deep networks using smoothing maximum technique. In: In *2022 IEEE/CVF conference on computer vision and pattern recognition (CVPR)*. New Orleans, LA, USA: IEEE; Jun. 2022. p. 784–93. <https://doi.org/10.1109/CVPR52688.2022.00087>.
- [30] Lofberg J. Yalmip: A toolbox for modeling and optimization in MATLAB. In: in *2004 IEEE international conference on robotics and automation (IEEE cat. No. 04CH37508)*. IEEE; 2004. p. 284–9.